Answer QUESTION 2 PARTS A and B on this page.

PART A

$r = 2\sin\theta^2$

$\frac{dr}{d\theta} = 4\sin\theta \cdot \cos\theta$ @ $\theta = 1.3 \approx$

$\boxed{\frac{dr}{d\theta} \approx 4\sin(1.3) \cdot \cos(1.3) \approx 1.031}$

PART B

$2\sin^2\theta = \frac{1}{2}$

$\sin^2\theta = \frac{1}{4}$

$\sin\theta = \frac{1}{2}$

$\theta = \pi/6, 5\pi/6$

$A = \frac{1}{2}\int_{\pi/6}^{5\pi/6} [2\sin^2\theta]^2 d\theta - \frac{1}{2}\int_{\pi/6}^{5\pi/6} [\frac{1}{2}]^2 d\theta$

$\boxed{A \approx 2.06677 \text{ units}^2}$

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

Answer QUESTION 2 PARTS C and D on this page.

PART C

$4\sin\theta\cos^2\theta - 2\sin^3\theta = 0$ @ $\theta = 0.95532 < \pi/2$

~~$\theta = 2.18628$~~ $> \pi/2$

~~$\theta = \pi$~~ $> \pi/2$

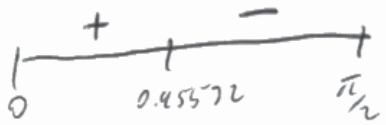
$x = 2\sin^2\theta\cos\theta$

candidate test

$x(0) \approx 0$

$x(0.95532) \approx 0.7648$

$x(\frac{\pi}{2}) \approx 0$



By the first derivative test, the distance x from the y-axis is at a max @ $\theta = 0.95532$ b/c $\frac{dx}{d\theta}$ goes from positive to negative, meaning x goes from increasing to decreasing, and $x(0.95532)$ is more distant on candidate test for $0 \le \theta \le \pi/2$

PART D

~~$\frac{dr}{dt} = \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}$~~ ~~$\frac{dr}{dt} = \sqrt{\frac{dx}{d\theta}^2 + \frac{dy}{d\theta}^2}$~~

$r = 2\sin^2\theta$

$\frac{dr}{dt} = 4\sin\theta \cdot \cos\theta \cdot \frac{d\theta}{dt}$

$\frac{dr}{dt}(1.3) = 4\sin(1.3) \cdot \cos(1.3) \cdot .15$

$\boxed{\approx 1.031 \cdot .15}$

Use a pencil or a pen with black or dark blue ink. Do NOT write your name. Do NOT write outside the box.

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